9 **(a)** **(i)** State what is meant by background radiation.

.....

..... [1]

(ii) State **one** source that makes a significant contribution to background radiation.

..... [1]

(b) Radioactive isotopes are used in some medical treatments.

Radium-223 ($^{223}_{88}Ra$) is an isotope of radium. Radium-223 decays by the emission of alpha (α) particles to an isotope of radon (Rn).

(i) Complete the nuclide equation for this decay.

$$^{223}_{88}Ra \longrightarrow {}^{\cdots\cdots}_{\cdots\cdots}Rn + {}^{\cdots\cdots}_{\cdots\cdots}\alpha$$

[2]

(ii) Radium-223 is injected into the body to treat a specific organ.

Explain why the source must be inside the body.

.....

..... [1]

(c) An isotope of technetium is injected into the body to detect cancer in one of the organs. The half-life of this isotope is 6 hours and it decays by emitting gamma (γ) radiation. The radiation is detected outside the body.

(i) Explain why a source of gamma radiation is used.

.....

..... [1]

(ii) Explain why a source with a short half-life must be used.

.....

..... [1]

[Total: 7]

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